

OpenDesign → Helix OTA Integration — GROUND TRUTH

Revision: 1 **Last modified:** 2026-07-09T00:00:00Z **Status:** active — supersedes the earlier guessed `OPENDESIGN_PLAN.md` **Scope:** How Helix OTA adopts OpenDesign (github.com/nexu-io/open-design) to design/refine its two React frontends (`clients/ota-manager`, `dashboard`) per Constitution §11.4.162 (mandatory OpenDesign UI design system) + §11.4.74 (extend-upstream) + §11.4.28 (decoupling). **Authority:** Evidence read directly from the operator-confirmed READ-ONLY clone at git 81b20dc (2026-07-09), tag/version `open-design@0.14.1`. Every claim below cites a file path or captured command output. Nothing here is guessed (§11.4.6 — items I could not verify are marked UNCONFIRMED:).

0. TL;DR (the real adoption model)

OpenDesign is **NOT** a CSS-token npm package and **NOT** a React component library you `npm install` into an app. It is a **local-first design PRODUCT** (pnpm monorepo, Electron desktop app + `od` daemon + web UI) that exposes itself to **coding agents over a stdio MCP server + `od CLI`**, and ships a large corpus of **design-system token packages** (153) + **skills** (163) + **plugins/templates** as **plain files** (`tokens.css`, `tailwind-v4.css`, `design-tokens.json`, `components.html`, `DESIGN.md`).

For Helix OTA (React19+Vite6+shadcn/Tailwind4 and React18+Vite), the adoption is a **combination**: 1. **Install the `od` daemon + wire the MCP plugin into the agent** (Claude Code) so the agent can read/generate designs (`od mcp install claude`), AND 2. **Consume the design-system TOKEN FILES directly** (`tokens.css` + `tailwind-v4.css` as `:root` custom properties + a Tailwind v4 `@theme`) as the project's light/dark design tokens — these are static, framework-agnostic files usable by BOTH Vite apps.

You do **NOT** consume @open-design/components — it is a private, workspace-internal, React-18.3.1-pinned package for OpenDesign’s OWN app (evidence §3).

Verdict: NEEDS-REVIEW before adoption (not blocked). Rationale in §6.

1. WHAT OpenDesign actually provides + its consumption model

1.1 It is a product/app, not a library

- package.json: "name": "open-design", "version": "0.14.1", "private": true, "packageManager": "pnpm@10.33.2", "description": "Local-first design product: detects your installed code-agent CLI, runs design skills + design systems, streams artifacts into a sandboxed preview.", "bin": { "od": "./apps/daemon/bin/od.mjs" }, engines: node ~24, pnpm >=10.33.2 <11.
- Workspaces (pnpm-workspace.yaml): packages/*, apps/*, tools/*, e2e. apps/ = daemon (Express+SQLite od server), web (Next.js), desktop (Electron), landing-page, packaged, telemetry-worker.
- Web-confirmed (github.com/nexu-io/open-design, fetched 2026-07-09): “open-source design application and platform — not a component library ... Not published as external npm dependencies — consumed as standalone applications or integrated via MCP protocol.” License **Apache-2.0** (LICENSE line 1).

1.2 Four consumption surfaces (README.md:300-395, QUICKSTART.md)

1. **Desktop app** (Electron) / **web UI** at http://localhost:7456.
2. **stdio MCP server** — od mcp install <agent> (16+ CLIs: claude, codex, cursor, copilot, gemini...). Installs ~/.config/<agent>/open-design.json; Claude Code gets a claude mcp add-json one-liner (README.md:389). The agent then reads skills/, binds a DESIGN.md, and emits a previewable <artifact>.
3. **od CLI** — od search-files, od get-file <path>, od get-artifact <slug>, od plugin run, od skill list, od plugin install (README.md:380-497). Requires a built daemon: od.mjs throws

Open Design daemon dist entry not found ... Run "pnpm --filter @open-design/daemon build" first. (captured — see §Verified).

4. **Docker** (deploy/, docker compose up -d).

1.3 The design-system TOKEN packages (the reusable asset for Helix)

design-systems/ holds **153** brand token packages (apple, ant, cal, linear, arc, bmw, canva, cisco...). Each is a self-contained folder emitting a canonical contract (evidence design-systems/apple/): - manifest.json — schemaVersion: "od-design-system-project/v1", declares the file set (design, tokens, designTokens, tailwind, components), preview/ pages, craft (color, accessibility-baseline). - tokens.css — the **machine-readable :root { ... } custom-property block** (56 standard tokens for apple). Header literally says agents must “paste the :root { ... } block verbatim into the first <style> of every artifact ... then reference everything via var(--name)”. - tailwind-v4.css — @import "tailwindcss"; @import "./tokens.css"; @theme { --color-*: var(--*) ... } — a ready **Tailwind v4 @theme** mapping over the tokens (matches Helix’s Tailwind 4 stack). - design-tokens.json — format: "od-design-tokens/v1", contract: "TOKEN_SCHEMA", per-token value/type/layer/confidence/source. - components.html + components.manifest.json — rendered component recipes/selectors. - system/kit.html + system/kit.dark.html — **light AND dark component kits** (both re-declare the token :root), plus preview/{colors, typography, spacing}.html. - DESIGN.md (+ ~20 i18n translations), USAGE.md (read-order + Do/Avoid).

The **cross-brand standard token schema** is packages/contracts/src/design-systems/ token-schema.ts (re-exported by design-systems/_schema/tokens.schema.ts). Standard token names include: --bg --surface --surface-warm --fg --fg-2 --muted --meta --border --border-soft --accent --accent-on --accent-hover --accent-active --accent-light --success --warn --danger --font-display --font-body --font-mono --text-xs...--text-4xl --leading-* --space-1...--space-22 --radius-xs...--radius-pill --elev-flat/ring/raised --focus-ring --motion-fast/base --ease-standard --container-max --section-y-* --container-gutter-*. This is the **component-level + theme token vocabulary** §11.4.162 refers to; brand switching = swap the :root block, keep the names.

1.4 Skills + plugins

- `skills/` = **163** `SKILL.md` folders (Claude Code SKILL convention + `od: frontmatter: mode/surface/platform/preview/design_system.requires/outputs`). Example `skills/8-bit-orbit-video-template/SKILL.md`.
- `plugins/` = MCP plugin + marketplace + registry + `spec/SPEC.md` (portable plugin contract: a dir with `SKILL.md` + optional `open-design.json`).
- `.claude-plugin/marketplace.json` — publishes the “open-design” Claude Code plugin (stdio MCP, “139 skills + projects/files/preview tools, served by your local `od` daemon. Requires `od` on `PATH`”).

Intended consumption model for an EXTERNAL React app: the agent (Claude Code) talks to the local `od` daemon over MCP to browse/generate/preview designs; the **durable, framework-neutral artifact you keep in your repo is the design-system’s `tokens.css` + `tailwind-v4.css` (+ `DESIGN.md` as design intent)**. OpenDesign emits HTML/CSS artifacts (“real HTML/CSS — drop it into Cursor/Codex/Claude Code to keep building as code”, `README.md:322`); it does not hand you a React component dependency.

2. Concrete adoption path for Helix OTA (§11.4.162)

§11.4.162 requires: install as a project dependency; use its `tokens/themes` (`light+dark`); component-level tokens; extend upstream for gaps; `light+dark` variants; no overlap. The **evidenced** path is a **combination of (a) agent/MCP wiring + (b) direct token-file consumption** — because that is the only model the artifact actually supports:

2.1 Author-time (agent-driven design) — MCP plugin + daemon

Exact commands (from `README.md:300-390`, requires Node ~24 + `pnpm` 10.33.2, both present on this host — captured `$Verified`):

```
# One-time, from the OpenDesign checkout (build the daemon the
  `od` bin needs):
corepack enable
pnpm install
```

```

pnpm --filter @open-design/daemon build      # od.mjs REQUIRES
      this (captured error proves it)

# Wire the stdio MCP server into Claude Code (agent used by
  Helix):
od mcp install claude                        # writes ~/.config/
      claude/open-design.json + claude mcp add-json snippet
# then, inside the agent:
#   > Use open-design to generate the ota-manager settings screen
      with the <chosen> design system

```

Per §11.4.161 rootless-container / §11.4.173 containerized-build posture, prefer the Docker path (deploy/, docker compose up -d, http://localhost:7456) over a bare pnpm install on the host when running the full product.

2.2 Build-time (what lands in the Helix repo) — direct token files

Pick a design system (e.g. design-systems/<brand>/), then vendor its **static** token files into each frontend and load them: - Files to copy (framework-agnostic, no OpenDesign runtime dep): design-systems/<brand>/tokens.css (the :root light+dark custom properties), design-systems/<brand>/tailwind-v4.css (the Tailwind v4 @theme), design-systems/<brand>/DESIGN.md (intent), components.manifest.json (recipes). - clients/ota-manager (React19 + Vite6 + Tailwind4 + shadcn): import tailwind-v4.css as the Tailwind entry (it already does @import "tailwindcss"), so shadcn components read --accent, --bg, --fg, --border etc. from the OpenDesign :root. - dashboard (React18 + Vite): import tokens.css for the :root custom properties; add a Tailwind v4 @theme (from tailwind-v4.css) if/when it adopts Tailwind, else reference var(--...) directly. - **Light+dark**: use system/kit.html (light) + system/kit.dark.html (dark) as the authoritative per-theme :root value sets and the visual golden for §11.4.170 host-rendered visual proof.

UNCONFIRMED: I have NOT yet read clients/ota-manager/ or dashboard/ (out of scope per guardrails), so the exact Tailwind entry filenames and current theme wiring in those two apps are unverified here; confirm before editing.

2.3 Why not “just npm install a package”

@open-design/components is "private": true, "version": "0.8.0", peerDependencies.react": "18.3.1", main: ./dist/index.mjs, consumed as workspace:* by the root and apps/* (evidence packages/components/

package.json). It is OpenDesign's INTERNAL app UI, not a distributable dependency, and it pins React 18.3.1 (conflicts with ota-manager's React 19). Do not depend on it.

3. §11.4.74 extend-upstream + §11.4.28 decoupling

- **Extend path (§11.4.74):** gaps are added the OpenDesign-native way — as a **new design-system folder** (its own manifest.json + tokens.css + tailwind-v4.css + DESIGN.md, validated by pnpm guard → check-design-system-manifests.test.ts) or a **new plugin/skill** per plugins/spec/SPEC.md (a dir with SKILL.md [+ optional open-design.json]). Apache-2.0 permits forking + upstream PRs. A **Helix-branded design system** (Helix OTA brand colors, light+dark, --accent = Helix brand) is the natural §11.4.162 “project brand colors from canonical assets” instantiation — authored as one new design-systems/helix-ota/ package.
- **Decoupling (§11.4.28):** OpenDesign is a third-party (org nexu-io), NOT an owned-org submodule, so §11.4.28(A) equal-engineering does NOT apply; it is a reusable, project-not-aware upstream. Two decoupled consumption options:
 1. **vendor the static token files** into Helix (no runtime coupling — cleanest for the two frontends), and/or
 2. **add it as a git submodule / pinned checkout** used only at author-time via the od daemon (never inject Helix-specific context into the OpenDesign tree). Nested owned-org chains are irrelevant (it is third-party). Recommend **(i) vendored token files as the shipped dependency + (ii) submodule/checkout only for the agent-time daemon**, keeping Helix runtime free of any OpenDesign code.

UNCONFIRMED: whether the operator wants OpenDesign as a tracked git submodule vs an external tool checkout — that is a §11.4.66 operator decision.

4. Safe-to-adopt verdict

NEEDS-REVIEW (adopt in a controlled feature stream — NOT blocked, NOT adopt-blindly-now). Reasons, evidenced:

- **GREEN signals:** Apache-2.0 (redistribution/fork/vendor OK); the token layer is plain static CSS/JSON with **native Tailwind v4 @theme output** matching Helix's Tailwind 4 stack; light+dark are first-class (kit.html/kit.dark.html); Node 24 + pnpm 10.33.2 already present on this host (captured); the design-token contract is stable + documented (TOKEN_SCHEMA).
- **REVIEW/RISK signals (why not “adopt now”):**
 1. Adopting the FULL product (daemon/desktop/web) is heavy: Electron, Next.js, SQLite, better-sqlite3/sharp/electron native builds (onlyBuiltDependencies), a pnpm install of a 422 KB-lockfile monorepo — must run under the §11.4.161 rootless / §11.4.173 containerized-build posture, not a bare host install.
 2. The od daemon binds 127.0.0.1 and does SSRF/model-endpoint gating (README.md:395) — a network service to security-review before running (§11.4.133 target-safety is about devices, but a new local daemon still needs review).
 3. @open-design/components React-18.3.1 pin conflicts with ota-manager React 19 — confirmed do-not-use, but a reviewer must ensure nobody wires it in.
 4. clients/ota-manager + dashboard current theme wiring UNCONFIRMED (not read) — must be inspected before choosing the token-import seam.
 5. §11.4.162 also mandates OpenDesign for **any** UI surface + visual-regression coverage (§11.4.170) — a real integration is a BIG feature-stream item (§11.4.167), not a drive-by edit.

Recommended next step: open a §11.4.167 feature work-stream feature/opendesign-adoption that (1) authors a design-systems/helix-ota/ brand package (light+dark), (2) vendors its tokens.css + tailwind-v4.css into both frontends behind the existing Tailwind entry, (3) wires the od MCP plugin for author-time only, (4) proves it with §11.4.170 host-rendered light+dark pixels. Hold the full-daemon/desktop adoption until the containerized-build + security review lands.

Verified evidence (commands run + key output)

- `node --version` → v24.18.0; `pnpm --version` → 10.33.2 (host meets engines: node ~24, pnpm >=10.33.2 <11).
- `git -C open-design remote -v` → `git@github.com:nexu-io/open-design.git`; `git log -1` → 81b20dc ... 2026-07-09 14:34:45 +0000 fix(web): avoid missing design-system preview assets (#5381).
- `cat package.json` → `name=open-design, version=0.14.1, private=true, bin.od=./apps/daemon/bin/od.mjs, description="Local-first design product..."`.
- `node apps/daemon/bin/od.mjs --help` → Error: Open Design daemon dist entry not found at .../apps/daemon/dist/cli.js. Run "pnpm --filter @open-design/daemon build" first. (proves od needs a build; clone left unbuilt/read-only).
- `cat packages/components/package.json` → `@open-design/components@0.8.0, private=true, peerDependencies.react=18.3.1, consumed workspace:* (internal)`.
- `ls design-systems | wc -l` → 153; `ls design-templates | wc -l` → 115; `ls skills | wc -l` → 163.
- `cat design-systems/apple/manifest.json` → `schemaVersion od-design-system-project/v1, files map (tokens: tokens.css, tailwind: tailwind-v4.css, designTokens: design-tokens.json, components: components.html), preview/ colors+typography+spacing`.
- `cat design-systems/apple/tailwind-v4.css` → `@import "tailwindcss"; @import "../tokens.css"; @theme { --color-accent: var(--accent); ... } (Tailwind v4 theme)`.
- `head design-systems/apple/tokens.css` → header instructs agents to paste the `:root { ... }` block verbatim; lint `apps/daemon/src/lint-artifact.ts` (raw-hex → P1, non-token accent → P0).
- `ls design-systems/apple/system` → `kit.html, kit.dark.html` (light + dark kits).
- `grep '"--..."' packages/contracts/src/design-systems/token-schema.ts` → the standard `TOKEN_SCHEMA` names (accent/bg/surface/fg/border/space/radius/text/elev/motion...).
- `.claude-plugin/marketplace.json` → publishes stdio MCP plugin "open-design" ("139 skills + projects/files/preview tools ... Requires od on PATH").
- `README.md:300-395` → `od mcp install <agent>`; Claude Code `claude mcp add-json`; artifacts preview at `http://localhost:7456`; "real HTML/CSS — drop it into Cursor/Codex/Claude Code"; read-only, 127.0.0.1-bound daemon, SSRF-gated.

- WebFetch github.com/nexu-io/open-design (2026-07-09): “design application and platform — not a component library ... Not published as external npm dependencies — consumed as standalone applications or integrated via MCP protocol”; Apache-2.0.
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Sources verified 2026-07-09

- <https://github.com/nexu-io/open-design> — repo README + About (fetched 2026-07-09): identity (design product/app, not a component lib), MCP/od/desktop/Docker consumption, Apache-2.0, pnpm monorepo, not published as external npm dep.
- Local READ-ONLY clone at `git 81b20dc / open-design@0.14.1` (commit dated 2026-07-09) — all file-path citations above.
- Constitution §11.4.162 (OpenDesign UI mandate), §11.4.74 (extend-upstream), §11.4.28 (decoupling), §11.4.161/§11.4.173 (rootless/containerized build), §11.4.167 (feature work-stream), §11.4.170 (host-rendered visual proof), §11.4.66 (operator decision), §11.4.6 (no-guessing).